EON

Efficiency Operation Nexus

A Bangla-native, multimodal intelligence that becomes a **tenant** inside a system that already exists — and turns its data into decisions, in the language its owner thinks in.

CONCEPT NOTE
PROTOTYPE · LIVE
2026

Project title	**EON — Efficiency Operation Nexus.** One brain, many bodies: an adaptive decision intelligence layer
Project stage	Working prototype, **deployed and running on a live production system and a live production database**
Sector	Applied AI · Enterprise & Government Digital Infrastructure · Bangla Language Technology
Team	Md Imran Hossain (Team Lead, CIS) · Nafisa Tabassum Nusrat (CIS) — Daffodil International University
Live system	`eon.gulfrabit.com`

01 Project summary

EON is an **adaptive intelligence layer** — not an application, not a chatbot, not an ERP feature. A portable reasoning core attaches to a system already running, **learns it from its own source and database**, and answers the people who depend on it — Bangla or English, voice or text, from live data, with **the arithmetic behind every figure** and the action it recommends next. Say yes, and it acts through the host's own screens.

The distinguishing decision: **EON computes its answers; it does not generate them.** No language model, no GPU, no per-query cost. Every sentence is arithmetic over a live database — **fast, reproducible, auditable, free to run**, and the data never leaves the server it lives on.

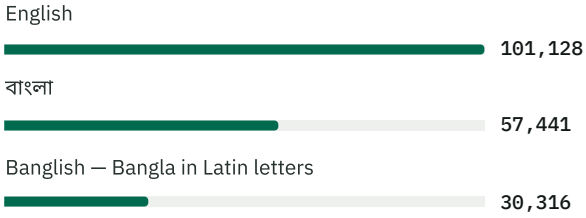
Proven in one demanding tenancy: a **twelve-company commercial ERP, 203 tables, 293 screens** — one of fifteen classes of Bangladeshi system with the same shape. What made the first work makes the next fourteen **a configuration problem, not a rebuild**.

1,191/1,191	120,105	15–60 ms	৳0
Director-level questions answered correctly, across three language surfaces	Distinct question-points the prototype can already address in one tenancy	Round trip from question asked to answer composed and spoken	Marginal cost per question — no model API, no GPU, no token meter

Engineering effort behind the prototype

Own source, excluding the host	~38,000 lines
Modules in the portable core	76
Reasoning tools reachable without a model	29 of 32
Host system mapped	1,376 routes · 293 screens
Data sections read, read-only	54

The question corpus, by language M



188,885 questions across 24 intents. The **1,191-question verified corpus** inside it answers at **100% in all three**; six of thirty Bangla answers still carry an English fragment, and all six are proper names the host stores in English.

What it is not

- **Not an ERP product.** The ERP is the first host, not the subject. The reasoning core has no ERP in it.
- **Not a wrapper on a foreign model.** Nothing is sent to an external endpoint. There is no endpoint.
- **Not a dashboard.** A dashboard shows a figure. EON explains why the figure moved and what to do about it.
- **Not a translation layer.** Bangla is not English output run through a translator — the reasoning itself runs in both languages.

The provenance rule of this document

Every quantity on the following pages carries its origin, because a concept note that mixes the two is not readable as evidence.

M **Measured** — observed on the running system by a repeatable test whose output is on file.

P **Projected** — derived from stated inputs. Every input is named, and changing it changes the result.

Nothing here is an unsourced market figure.

02 Problem statement

Bangladesh has spent two decades successfully installing software. Ministries, banks, hospitals, factories and SMEs now run systems that record faithfully and at scale. **The national problem is no longer recording.** It is that recorded data does not become a decision without a trained human between the two — and there are **not enough of them**.

GAP 01 — DECISION

The data is present; the decision is absent

A system holds every figure a decision-maker needs and **prints none of the reasoning**. No screen says why profit fell, only that it did. No report ranks twelve companies, or qualifies itself when the data is incomplete. **That gap is crossed manually, or not at all.**

GAP 02 — LANGUAGE

The system speaks English; the country thinks in Bangla

The person accountable reasons in Bangla; every screen and report is in English. **Machine translation does not close this**: translated output carries English reasoning underneath, gets the numeric scale wrong (লাখ, কোটি), and cannot be read aloud — most machines have no Bangla voice at all.

GAP 03 — SOVEREIGNTY

The obvious fix exports the nation's data

The default way to add intelligence is to send the system's contents to a foreign model API. **A ledger, a patient list or a land record then leaves the country on every query**, at a foreign-currency cost that scales with usage — and no answer can be audited, because the same question returns a different sentence tomorrow.

GAP 04 — INTEGRATION

Adding intelligence usually means replacing the system

Forking, migrating or replacing the host each risks a working system, consumes a budget cycle, and breaks at the next upgrade. **Institutions correctly refuse.** So the systems most in need of an intelligence layer are the least able to accept one.

03 The proposed solution — one brain, many tenancies

EON closes all four with a single architectural idea: **the intelligence does not own the system it serves — it rents a room in it.**

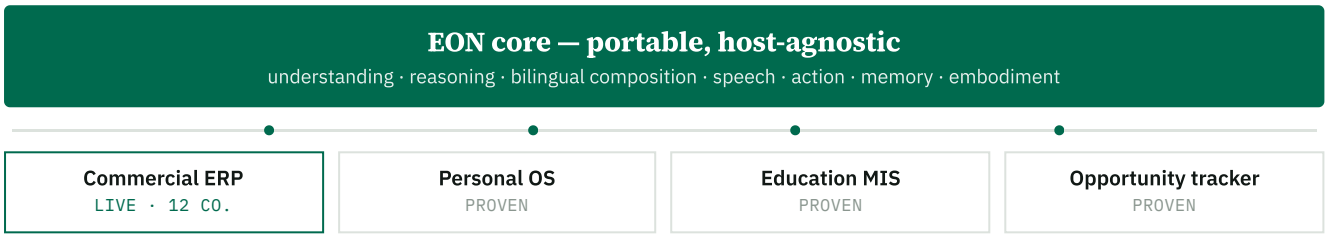
CLOSES	HOW, CONCRETELY
Decision gap	An invariant loop: `read live data → decide what matters → detect risk and waste → rank the next action → explain the rule → act, on confirmation, reversibly.` **It does not change between hosts**; only what it reads does.
Language gap	Bangla and English are **parallel composition targets**, not source and translation. Each answer is written natively in the language asked, on the Bangladeshi numeric scale, and rendered to speech server-side — so **a machine with no Bangla voice still speaks Bangla**.
Sovereignty gap	**No model API. No GPU. No outbound call.** The reasoning runs in PHP and SQL beside the host database, and the same question returns the same sentence every time because it is arithmetic.
Integration gap	EON attaches by **injection, not modification**: one script tag appended by an output-buffer callback. `M` **Zero lines of host source are edited**; removing two files returns the host to exactly what it was.

> **Why this is the interesting claim**
>
> Most AI systems are judged on how well they generate. **EON should be judged on the opposite: it cannot invent a number.** Where two figures genuinely conflict it **reports both and names the gap** rather than quietly choosing one — and that habit of **qualifying its own answer** is the difference between a dashboard and an adviser.

04 How EON becomes a tenant of a system it has never seen

A tenancy is established in four moves, none of which requires the host's cooperation beyond read access.

#	MOVE	WHAT HAPPENS	PROVEN AT
1	**Survey**	A mapper reads the host's own source and schema and writes a machine-readable map — **every route, screen, controller, model and table**.	1,376 routes 203 tables
2	**Discover**	Schema auto-discovery types every column, detects date and deadline fields by name and by sampled value, and picks a human label per entity. **No manual field list.**	54 sections 661 records
3	**Attach**	The companion is injected into every host page. Sign-in is **inherited**: the encrypted session cookie is decrypted and verified, so being signed into the host is being signed into EON.	0 host files edited
4	**Reason**	The portable core — understanding, answering, phrasing, speech, action — **runs unchanged**. Only the adapter and knowledge map are tenancy-specific.	45 subjects 16 aspects



One core, four bodies. M Three earlier systems were merged into the portable core — a real three-way merge against the fork point, not a copy. **Every adapter beneath the rail is replaceable; nothing above it changes.**

Deliberately polyglot — each layer in the language that suits it

LAYER	LANGUAGE	WHY THERE
Reasoning, data access, API	PHP 8.3	Beside the host, same server, same session, no bridge to secure
Set computation	SQL	Aging, trial balance and roll-ups belong in the database
Heavy analytics	Python	Forecasting and file export — optional, never on the critical path
Face, voice, presence	JavaScript	Embodied companion, speech in and out, offline fallback

Security is inherited, not invented

- The database user is **read-only**.
- Every endpoint is **fail-closed** — no session, no data.
- Every write goes back through **the host's own form and its own permission check**. Nobody can ask EON for what the host would refuse them.
- Money movements and deletions **always** require confirmation; a half-filled form stops rather than guessing.
- A refusal from the host is read and reported as a refusal — a success code is not taken as success.

Multimodality, as built

MODALITY	STATE IN THE PROTOTYPE
Text	**Live.** Any phrasing, three language surfaces, formal or colloquial, question or order.
Speech in	**Live.** Push-to-talk and hands-free wake word, English and বাংলা.
Speech out	**Live.** Rendered server-side, so **Bangla is spoken correctly on a machine with no Bangla voice** — figures said the way a person says them, not digit by digit.
Action	**Live.** Instruction → proposal → confirmation → posted to the host's own route.
Embodied presence	Built — a real-time 3-D companion with physics, pathfinding, emotion and personality engines. Present in the earlier bodies; **not yet wired to the enterprise core**.
Document intake	Built in an earlier body — reads a PDF, spreadsheet or CSV with **zero configuration**. Not yet carried into the enterprise tenancy.

05 The path of a question

Nothing is queued and nothing is transmitted. The entire round trip happens on the host's own server.

01	02	03	04	05
Ask	**Understand**	**Answer**	**Phrase**	**Speak**
Voice or text. Bangla, English, or Banglish written in Latin letters.	Intent and language are read; anything unrecognised is parsed structurally instead.	The ledger is read, the rule applied, the figure decomposed until it is actionable.	Composed natively in the language asked, on the Bangladeshi numeric scale.	Rendered to audio server-side and cached, so a repeat answer is instant.

It composes rather than enumerates

EON holds **no list of supported questions**. It scores the idioms people actually use; anything falling through is read **structurally** — a verb pointed at a subject, or a named record paired with an aspect.

So the capability surface is large without being hand-written: **45 subjects × 16 aspects × phrasings × 2 scripts** compose to `M` **120,105 addressable question-points**, a 1,500-point stratified sample of which answers at **100%**.

It reasons about a figure until it is actionable

> Asked **"why is cash so tight?"** — composed, not retrieved:
>
> "Cash ৳1.89 L against a burn of ৳2.17 L a month — about 0.9 months, and **99% of that burn is payroll**. ৳15.5 L is owed to you and all of it is past due. Against that you owe ৳20.7 L. So the gap is ৳17.8 L, and **collecting every overdue taka would still not close it.**"

That last clause is what a decision-maker needs, and **no report in any host system prints it**. It is reached by decomposition — which of income, direct cost or overhead moved, and what is unapproved and therefore absent.

06 Where EON does not compromise

Six figures that **do not move with scale, usage, or the sensitivity of the data. None is a policy commitment someone must audit and enforce** — each is a property of the architecture: the thing that would break it does not exist in the system.

৳0 **marginal cost per question** No token meter and no per-seat fee. The only charge is installation.	**0** **bytes that leave the server** Not a promise about where data is sent — there is no outbound call to make.	**0** **GPUs or model APIs** The reasoning is arithmetic in PHP and SQL, on ordinary shared hosting.
0 **lines of host source edited** Two injected files. Delete them and the host is exactly what it was.	**0** **numbers that can be invented** There is no generative step in which a figure the database does not support could appear.	**0** **variance between two answers** The same question returns the same sentence, and any answer can be re-derived.

And five differences the zeros do not capture

PROPERTY	THE CONVENTIONAL APPROACH	EON
Source of an answer	A model summarising exported data	Arithmetic over the live database — **it cannot invent a number**
Latency	Network round trip to a foreign region	`M` 15–60 ms, same machine
Bangla	Translated output over English reasoning	Reasoned and composed natively in both languages
Scope of action	Read-only conversation	Executes through the host's own forms, permissions intact
Failure mode	A confident, plausible, wrong paragraph	An honest "I do not know" — the cheaper failure by far

> **The trade being made, stated plainly**
>
> Because nothing is generated, a question outside the modelled grammar gets a refusal rather than a plausible paragraph — a real limitation, and **the correct one for ledgers, patient records and land registers**, where a confident wrong answer costs far more than no answer. A language model can be added later **for phrasing**; it is not needed for truth.

07 What is measured M

Every row below is a repeatable test with its output on file. Each suite is built to **print its failures**, not to produce a grade.

WHAT IS TESTED	SIZE	RESULT
Corpus generated and counted	290,368 records, counted line for line against the manifests	290,368
Director evaluation corpus	1,191 questions, all three languages	100%
Whole-system sweep	242 questions and orders, full chain, real browser	242 / 242
Verb × subject grid	435 combinations, both scripts	100%
Record × aspect grid	120,105 points; sample of 3,000	2,999 / 3,000
Name resolution across scripts	22 cases — a Bangla name reaching a Latin record	22 / 22
Action path — task, message, notice	36 cases, end to end, both languages	36 / 36
Action path — money	87 cases against the host's real payment routes	87 / 87
Narrowing a question, and comparing two periods	41 cases — the refusal and the false positive	41 / 41
File export, opened as a real workbook	26 cases — unzipped, all five XML parts parsed	26 / 26
Access audit, both directions	33 checks — all needed answer, all private refuse	33 / 33
Live end-to-end, including audio	10 questions on the running site, audio verified	10 / 10
Live deployment check	34 assertions against the production site	34 / 34
Accessibility	Unnamed controls, unlabelled inputs, contrast, Bangla marked	0 failures

A question passes only if it lands on the right intent, avoids the fallback, is **about the thing that was asked**, and answers in the language asked. **Three of four is a failure.**

08 What it found in the live books, unprompted

Reading a real production database, EON reported four things nobody had raised. **None is a software fault; all four are money.**

FINDING	CONSEQUENCE
৳**5.27 L** of a month's sales never reached the ledger — invoiced ৳9.1 L, ledger carries ৳3.8 L	Every profit figure understated the business by more than half
23 of 80 payslips do not add up; the worst is ৳30,000 out	Direct payroll exposure, repeating every month
Advances not applied — a customer paid ৳**9.98 L** and an invoice still showed ৳8.5 L due	Collections chasing money already banked
Running balances stored in entry order rather than date order	A back-dated entry leaves the ledger disagreeing with itself

The published gaps — three closed, two open

A prototype that hides its edges cannot be evaluated. This list was published before anyone asked. **Three entries are now closed and measured** (24 Aug); the rest stand.

- **Closed** — **a question that narrows.** The phrase is resolved against dimensions the data really carries; "in Chittagong" is **refused**, because no city column exists in the host.
- **Closed** — **comparison across periods.** Cash and headcount are positions, not flows, so comparing them is **declined out loud**.
- **Closed** — **file export.** The workbook is written in plain PHP — no library, no binary, **no network call**.
- **Still open — no free-form drafting** beyond structured notices and reminders.
- **Still open, deliberately — rules are a calendar, not a rate card.** VAT and TDS percentages are the accountant's to supply; **a guessed rate is worse than an honest refusal**.

09 The headroom model — why the prototype is roughly 1/900th of the system P

"Nine hundred times" is not a slogan: it is **the product of five independent axes**, each with a measured value today and a ceiling the architecture already admits. **Every factor is an input — change any of them and the product changes with it.**

AXIS	TODAY (MEASURED)	CEILING THE ARCHITECTURE ALREADY ADMITS	FACTOR
Host tenancies	1 live, 3 more proven	15 classes of Bangladeshi system with the same shape (page 7)	**15×**
Subjects per tenancy	45 subjects, 16 aspects	180 — full coverage over a large host schema	**4×**
Modalities	3 wired (text, voice, action)	6 — adding embodied presence, document intake and visual input, all built in earlier bodies	**2×**
Language surfaces	2 natively composed	5 — regional Bangla variation and one export language	**2.5×**
Autonomy tiers	1 — ask, propose, confirm, act	3 — adding continuous watch, then policy-bound autonomous execution	**3×**
Compound headroom	15 × 4 × 2 × 2.5 × 3		**900×**

The prototype already addresses M **120,105** question-points in one tenancy; across the five axes the same core addresses a projected P **≈108 million** — not because more questions were written, but because composition multiplies. The mechanism is measured, not hoped for: when structural reading was added, the verb × subject grid moved from **68.3% to 100%** with **no new question written by hand**.

10 It arrives already knowing the kind of system it is joining

Until now EON learned a host only by surveying it. That answers about this system; it does not arrive understanding what **kind** of system it is. **Two generated corpora now carry that understanding** — what a new host inherits on day one. Every business type is crossed with every module it really runs, and each fact is written in that trade's own word: a hospital reads "patient" where a university reads "student".

290,368	101,483	188,885	24
records across the two corpora M	domain knowledge contexts, one per fact	question scenarios, every way a person asks	business types, each crossed with its own modules

THE 24 BUSINESS TYPES THE CORPUS COVERS

University · school · hospital · pharmacy chain · manufacturing plant · garments factory · textile mill · printing press · automotive workshop · furniture workshop · construction company · real estate developer · retail chain · e-commerce · trading and import house · restaurant · hotel · travel agency · IT services firm · internet service provider · logistics company · microfinance institution · NGO · agro and poultry farm

Built from **86 module packs** — 24 every business runs, 62 only some trades do — crossed into **26 facets**: entities and their fields, processes and their steps, measures and how to read them, reports, roles, permissions, traps and their fixes, posting rules. The questions span **24 intents**, from "what is this number" to "how do I fix it". **No trade is a stub**, and the generator is deterministic — same inputs, same ids, every run.

What host number two actually costs P

Projected from the first tenancy, where the mapper, the discovery layer and the whole portable core did not yet exist and were built alongside it.

STAGE FOR A SECOND HOST	PROJECTED	BASIS
Survey and map the host automatically	**hours**	the mapper is written and ran against 1,376 routes
Adapter and knowledge map for the new schema	**2–4 weeks**	the only genuinely tenancy-specific work
Attach, inherit session, verify access both ways	**days**	injection and the 33-check audit are both reusable
Reasoning, bilingual composition, speech, action	**unchanged**	**host-agnostic by construction**

Built — and precise about what is not yet claimed

Both corpora are generated and counted against their own shards: **101,483 domain knowledge contexts and 188,885 question scenarios — 290,368 records across 24 verticals**, in English, বাংলা and Banglish. What they do not yet carry is a pass rate: the failure-printing runner behind page 5 must be pointed at them first. **Building a corpus and proving it are two different pieces of work, and this document does not merge them.** Note that 24 business types already exceeds the 15 the headroom model assumes, making that projection conservative.

11 The fifteen tenancy classes

Each already runs software that records faithfully and decides nothing. Each has an accountable person who reasons in Bangla. **That combination is exactly the shape EON is built for.**

TENANCY CLASS	WHAT EON WOULD ANSWER THERE	FIT
SME & enterprise ERP	Cash position, leakage, payroll errors, collections, next action	`proven live`
Government service portals	Backlog by desk, applications past the service standard, where a file is stuck	`direct`
Local government desks	Union and upazila caseloads, entitlement queries in Bangla by voice	`direct`
Land records and registry	Conflicting entries surfaced and **both** reported; chain-of-record	`direct`
Health facility systems	Stock-out risk, referral backlog, staffing against attendance	`direct`
Education MIS	Attendance, results, at-risk students, teacher workload	`proven earlier`
Microfinance and MFS	Portfolio at risk, repayment ageing, agent float	`direct`
RMG floor and compliance	Line efficiency, order ageing, audit-readiness, wage integrity	`direct`
Freight, port and logistics	Consignment ageing, demurrage exposure, document completeness	`direct`
Agriculture and subsidy	Disbursement coverage, duplicate beneficiaries, seasonal timing	`direct`
Utility billing	Collection efficiency, meter anomalies, arrears ageing by area	`direct`
Tax and VAT desks	Return completeness, filing calendar, declared against recorded	`partly built`
Bank back office	Reconciliation breaks, suspense ageing, exception queues	`direct`
NGO programme management	Budget burn against plan, coverage, donor deadlines	`direct`
Telecom and ISP operations	Fault ageing, SLA breach risk, field workload distribution	`direct`

National alignment

- **Data sovereignty.** No record leaves the server — a property of the architecture, not a promise to enforce.
- **No foreign-currency dependency.** No token meter, no GPU rental. **The only charge is installation.**
- **Bangla as a first-class computing language** — correct numeric scale, spoken on machines with no Bangla voice.
- **Reuse of public investment.** Systems already paid for become more capable without being replaced.

12 Current status

Stage	**Deployed, reading a live production database**
Tenancies	One live across twelve companies; three earlier bodies merged
Operating cost	Shared hosting. No model API, no GPU
Known gaps	Published in full on page 5
Next milestone	**A second tenancy in a different sector** — "portable by design" becomes "portable, demonstrated"

13 Direction of development

1	**Close the published gaps**	Comparison across periods; a segment filter that refuses rather than widening; a stated forecast; file export
2	**Measure the corpora**	Point the failure-printing runner at all 100,000 question scenarios; publish coverage per family and language
3	**Second tenancy**	Adapter and knowledge map for a non-ERP host; re-run every suite unchanged; publish what carried across
4	**Full multimodality**	Carry embodied presence and document intake into the enterprise core; add visual input
5	**Autonomy under policy**	Continuous watch, then bounded autonomous action under a written policy, with full audit trail
6	**Tenancy toolkit**	Package survey, discovery, adapter scaffolding and audit suites so a third party can establish a tenancy

Risks, pre-answered by the design: a host upgrade cannot conflict (nothing in the host is edited); a wrong answer cannot be generated (nothing is generated); data cannot leak (no outbound call); an unwanted action cannot fire (money and deletions always confirm); and coverage cannot look complete when it is not (every suite prints its failures).

An institution's software already knows everything. EON is the layer that makes it say so — in Bangla, on its own server, and with the arithmetic attached.

One tenancy of a system built to hold many. It is **running today**, **measured rather than demonstrated**, and it **publishes what it cannot do**. What it needs next is not a larger model — it is a second host.

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Live system
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Prototype · running on production data